Introduction to Cybersecurity

Introduction



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Speaker



University Professor

Since 2011

CISO | Head of SecOps

Since 2019

Awards

- Microsoft SQL Server MVP
- Microsoft Data Platform MVP
- Microsoft Security MVP

Since 2009

MVP

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Professional

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Summary

- Leak statistics
- Introduction to Security



World's Biggest Data Breaches & Hacks

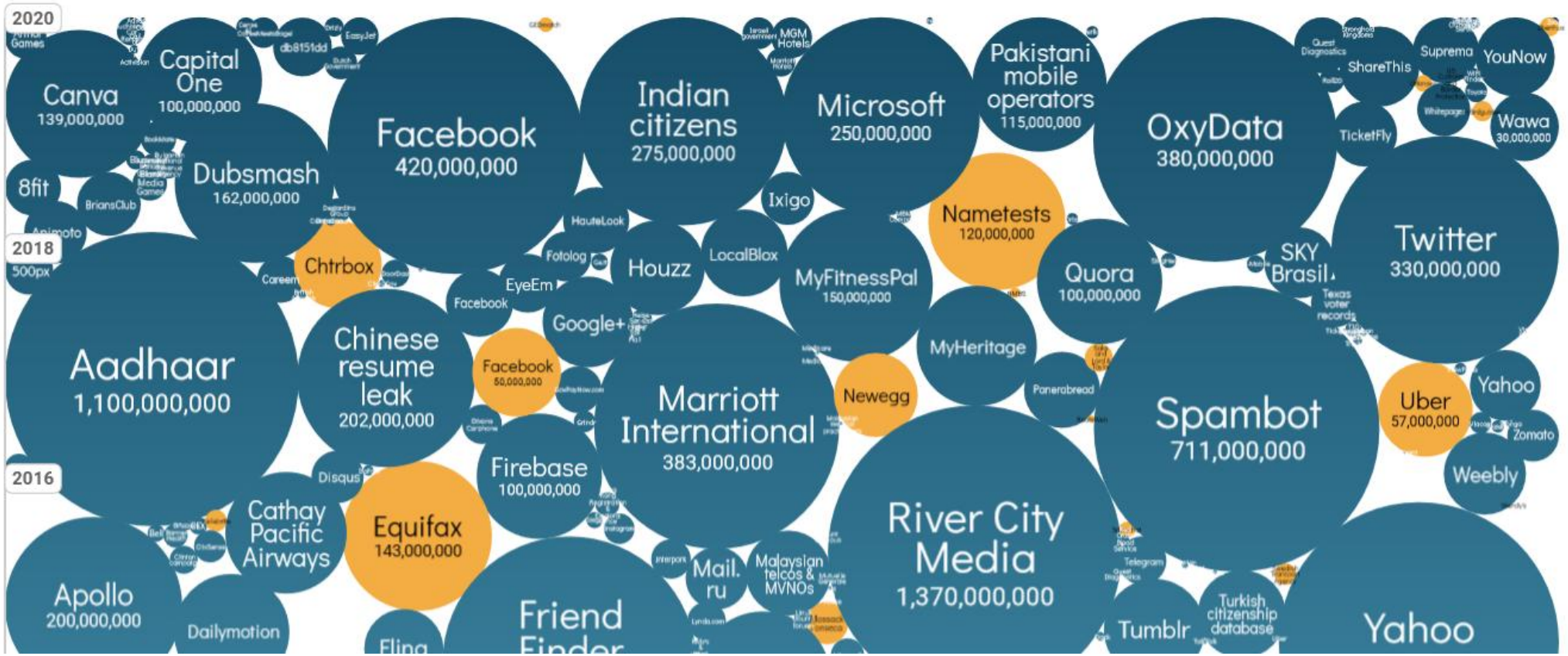
interesting story

Selected records over 30,000 records

updated: Nov 2020

size: records lost **filter**

search...



Marriott Hacking Exposes Data !



Marriott International acknowledged on Friday that an "unauthorized party had copied and encrypted information" belonging to about 500 million customers on its Starwood reservations system.
Mauritz Antin/EPA, via Shutterstock

We have a winner !!!

Yahoo confirms major breach — and it could be the largest hack of all time

Kif Leswing Sep. 22, 2016, 2:38 PM



AP Photo

Yahoo on Thursday revealed a massive data breach of its services.

Turkish Citizenship Database Leak !

Apr
03
2016

Turkish Citizenship Database Leak (Update 2)

Posted by Dissent at 2:44 pm Breach Incidents, Exposure, Government Sector, Hack, Non-U.S.

Seen online after a subsequently-deleted tweet called attention to it:

This [redacted] 6.6 GB file, purportedly containing clear-text information on **49,611,709** Turkish citizens, including the following details:

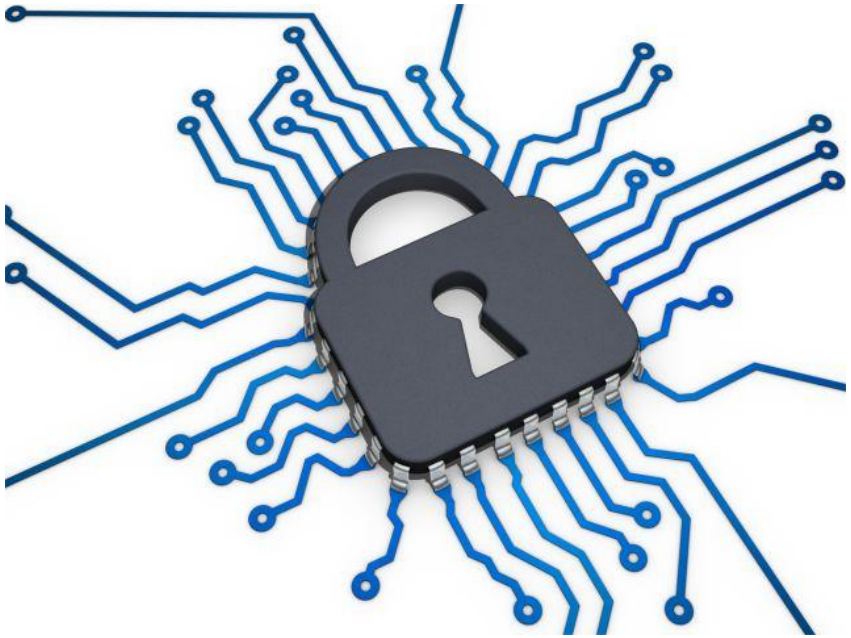
- National Identifier (TC Kimlik No)
- First Name
- Last Name
- Mother's First Name
- Father's First Name
- Gender
- City of Birth
- Date of Birth
- ID Registration City and District
- Full Address

Ashley Madison..EPIC..leak



Security ?

- Cyber security
  - Information securtiy
  - Network security
  - Computer security
  - IoT
  - …



...it is a process not final state !!!!!

There is no 100% secure system.

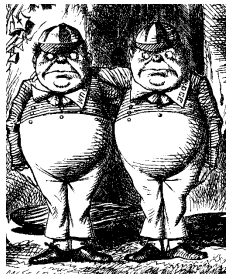
The Cast of Characters

- Alice and Bob are the **good guys**



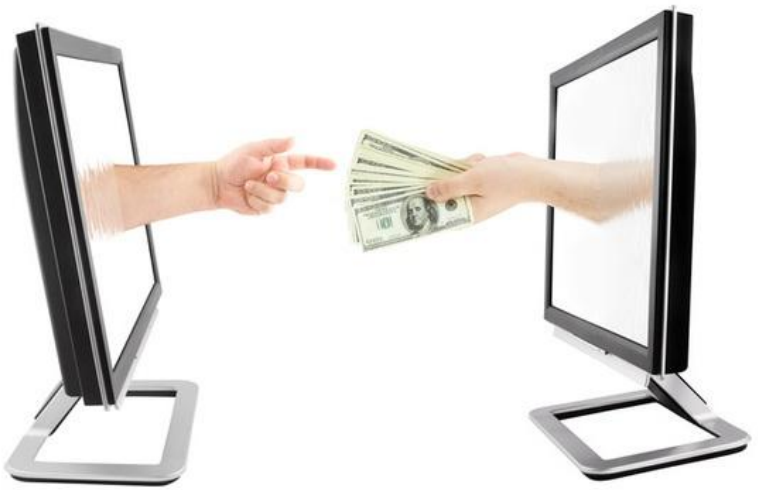


- Trudy is the **bad** **“guy”** 



Alice's Online Bank

- Alice opens Alice's Online Bank (AOB)
- What are Alice's security concerns?
- If Bob is a customer of AOB, what are his security concerns?
- How are Alice's and Bob's concerns similar?
  - How are they different?
- How does Trudy view the situation?



CIA…



- CIA == Confidentiality, Integrity, and Availability
- AOB must prevent Trudy from learning Bob's account balance
- **Confidentiality:** prevent unauthorized *reading* of information
  - Cryptography used for confidentiality



…CIA…

- Trudy must not be able to change Bob's account balance
- Bob must not be able to improperly change his own account balance
- **Integrity:** detect unauthorized *writing* of information
  - Cryptography used for integrity



…CIA

- AOB's information must be available whenever it's needed
- Alice must be able to make transaction
  - If not, she'll take her business elsewhere
- **Availability:** Data is available in a timely manner when needed
- Availability is a "new" security concern
  - Denial of service (DoS) attacks



CIA issues: Crypto

- How does Bob's computer know that "Bob" is really Bob and not Trudy?
- Bob's password must be verified
  - This requires some clever **cryptography**
- What are security concerns of pwds?
- Are there alternatives to passwords?



CIA issues: Protocols

- When Bob logs into AOB, how does AOB know that “Bob” is really Bob?
- As before, Bob’s password is verified
- Unlike the previous case, **network** security issues arise
- How do we secure network transactions?
  - **Protocols** are critically important
  - Crypto is important part of protocols



CIA issues: Access Control

- Once Bob is *authenticated* by AOB, then AOB must restrict actions of Bob
  - Bob can't view Charlie's account info
  - Bob can't install new software, etc.
- Enforcing these restrictions: *authorization*
- **Access control** includes both authentication and authorization



CIA issues: Software

- Cryptography, protocols, and access control are implemented in **software**
  - Software is foundation on which security rests
- What are security issues of software?
  - Real world software is complex and buggy
  - Software flaws lead to security flaws
  - How does Trudy attack software?
  - How to reduce flaws in software development?
  - And what about malware?



The People Problem...

- People often break security
  - Both intentionally and unintentionally
  - Here, we consider the unintentional
- For example, suppose you want to buy something online
  - To make it concrete, suppose you want to buy *Information Security: Principles and Practice*, 2nd edition from amazon.com



…The People Problem…

- To buy from amazon.com…
  - Your Web browser uses SSL protocol
  - SSL relies on cryptography
  - Access control issues arise
  - All security mechanisms are in software
- Suppose all of this security stuff works perfectly
  - Then you would be safe, right?



…The People Problem

- What could go wrong?
- Trudy tries man-in-the-middle attack
  - SSL is secure, so attack doesn't "work"
  - But, Web browser issues a warning
  - What do you, the user, do?
- If user ignores warning, attack works!
  - None of the security mechanisms failed
  - But user unintentionally broke security



Think Like Trudy...

- In the past, no respectable sources talked about “hacking” in detail
  - After all, such info might help Trudy
- Recently, this has changed
  - Lots of books on network hacking, evil software, how to hack software, etc.
  - Classes teach virus writing, SRE, etc.



…Think Like Trudy…

- Good guys **must** think like bad guys!
- A police detective…
  - …must study and understand criminals
- In information security
  - We want to understand Trudy's methods
  - Might think about Trudy's motives
  - We'll often pretend to be Trudy

…Think Like Trudy…

- Is all of this security information a good idea?
- Bruce Schneier (referring to *Security Engineering*, by Ross Anderson):
  - “It’s about time somebody wrote a book to teach the good guys what the bad guys already know.”

…Think Like Trudy

- We must try to think like Trudy
- We must study Trudy's methods
- We can admire Trudy's cleverness
- Often, we can't help but laugh at Alice's and/or Bob's stupidity
- But, we **cannot** act like Trudy
  - Except in this class…

In This Course…

- Think like the „bad guy“
- Always look for weaknesses
  - Find the ***weak link*** before Trudy does
- It’s OK to break the rules
  - What rules?
- But don’t do anything illegal!



